

Improving mechanical performance for extreme environments

Project Solution Approach

WSU Mechanics and Materials



WSUM-PythonApps

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TABLE OF CONTENTS

I. Introduction	3
II. System Overview	3
III. Architecture Design	3
III.1. Overview	3
III.2. Subsystem Decomposition	4
III.2.1. Image Segmentation	4
III.2.1.1. Description	4
III.2.1.2. Concepts and Algorithms Generated	4
III.2.1.3. Interface Description	4
III.2.2. Data Analysis	5
III.2.2.1. Description	5
III.2.2.2. Concepts and Algorithms Generated	5
III.2.2.3. Interface Description	5
III.2.3. View/Controller	6
III.2.3.1. Description	6
III.2.3.2. Concepts and Algorithms Generated	6
III.2.3.3. Interface Description	6
IV. Data Design	7
V. User Interface Design	7
VI. Glossary	7
VII. References	8
VIII. Appendices	8

I. Introduction

The purpose of this design document is to illustrate the fundamental design structure of our image analysis project in collaboration with the Washington State University Material Science Research Laboratory. We will be focusing on the three main design elements: Architecture, Data, and User Interface. In Architecture Design, we will dive into the overall structure of our software through the different subsystems and give a more in-depth explanation of the top-level design view of our system. Within this section we will also focus on the subsystem's purpose, functionality, and connected interfaces. Lastly, for both the data and User Interface design we will have detailed explanations and example illustrations on their usage and components in the project.

Additionally, this section aims to create a roadmap for both developers and stakeholders to understand the design choices and process flow as development for this project goes on, alongside allowing for comparisons to intended objectives and desired outcomes.

Our goal is to develop a program for materials researchers that will analyze images of the materials used to create protective shields that have been damaged due to circumstances found in extreme environments, and then gather information about their breakage patterns. The main technique that we will be using to distill this information is image segmentation. This will allow us to format and process the images in such a way that we can efficiently take various measurements and further analyze them. By analyzing the damage as well as imperfections in the materials, we will be able to better learn which circumstances allow for the best performance. This information will ideally be used in the development of stronger and more durable materials.

II. System Overview

The core functionality of our project is the automated segmentation and analysis of images with high accuracy, as this program will be compared with traditional manual calculations. The functionalities can be separated into three main components: Image Segmentation, Statistical Analysis, and a Graphical User Interface. For design methodology we decided on the Scrum Sprint method since we are using Python and MATLAB for development. We also decided on this methodology since the 421 class structure is based on the Scrum Sprint model. Regarding the design of our data structures, there was no concrete determination for how it should be structured. Our current plan is to implement primarily local and temporary variables in Python and MATLAB. However, depending on the needs of the project this may change. While our current situation allows our program to be easily adjustable, future implications will have to be considered as we near certain project breakpoints, such as when integrating the UI and image database with the Python package PyInstaller.

III. Architecture Design

III.1. Overview

The WSUM team has chosen to use a sub-style of the Call and Return architecture, namely the Object-Oriented architecture Model (Pranathi) for this project. Since the project is implemented in python and MATLAB, the architecture model that conformed the most with python was the Object-Oriented model. Below is a component diagram that illustrates the overarching

architecture of our program. The purpose of this diagram is to first have a clear diagram that allows our team to neatly implement each component of our system. In addition, it provides a representation of the component architecture that is decomposed in the next section. As seen in the diagram below, the Python UI will communicate with select system handlers. Some handlers that the Python UI element would communicate with are the UI handler to set up the page layout, icons, any other dynamic and static elements.

III.2. Subsystem Decomposition

III.2.1. Image Segmentation

III.2.1.1. Description

The image segmentation component of this application will be responsible for all the image processing that this application will handle. Loaded images will all be segmented to allow efficient analysis of the individual pores. Once segmented, labels will be applied to the segmented portions and data regarding the dimensions and aspects of the pores will be collected and stored with respect to their labels.

III.2.1.2. Concepts and Algorithms Generated

The methods of image segmentation that we will be using are watershed and adaptive thresholding. Since we have two different use cases, segmenting pores and segmenting imperfections, we plan on using those two methods to segment the use cases respectively. For the pores, since each pore is likely to be sharing a border with another one, thresholding would not work as well. Once the threshold is applied many of the pores will meld together and the segments will be inaccurate. To account for this, we will be using the watershed approach. This algorithm will find peaks and valleys throughout the image based on the coloring, and once the valleys are determined they will begin to fill up with different colors of “water.” By using different colors to fill each valley, it will be easy to differentiate separate pores regardless of how close they are to one another. When it comes to more general imperfections in the images, thresholding will work much better than watershed. For imperfections, they very rarely share borders with each other, so by thresholding the image it will efficiently segment the individual imperfections without the risk of overlap. Furthermore, the use of adaptive thresholding will base the thresholds off the local variance in color around the imperfections, so even if there is a large range in intensity throughout the image the individual segments will still be properly applied.

III.2.1.3. Interface Description

Services Provided:

Service Name	Service Provided To	Description
Information Gathering	Data Analysis Component	When the images are segmented data regarding the dimensions and other aspects of the segments are acquired. This information will be used in the data analysis component.

Segmented Images	Data Analysis Component	The images will be segmented and supplied to the data analysis component for the user to view the images pre and post segmentation.
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Services Required:

Service Name	Service Provided From
Image Importing	View/Controller

III.2.2. Data Analysis

III.2.2.1. Description

The data analysis component is responsible for analyzing the data that is collected from the image segmentation. Once data is collected, it will be formatted and manipulated to visualize what has been found and provide insightful meaning.

III.2.2.2. Concepts and Algorithms Generated

The analysis will be done using pandas, NumPy, and Matplotlib, three very powerful Python libraries made for mathematical calculations, statistical analysis, and data visualization. The data will be primarily formatted using pandas DataFrames. This data structure allows the data to be put into a table that can be easily manipulated using a variety of operations. NumPy will be used for the mathematical portion of the analysis. We plan on using it to find the variance in the data as well as any other insight requested by the client. Once all the information is formatted and computed, Matplotlib will be used to display it such that it can be easily understood by the application user.

III.2.2.3. Interface Description

Services Provided:

Service Name	Service Provided To	Description
Data Visualization	View/Controller	Generates plots for each image that show the image pre and post segmentation, as well as the visualizations of the analyzed data corresponding to each image.

Services Required:

Service Name	Service Provided From
Information Gathering	Image Segmentation Component
Segmented Images	Image Segmentation Component

III.2.3. View/Controller

III.2.3.1. Description

The View/Controller component will be the main interface between the user and the application. On top of being the UI, this component will also control the flow of the application, bridging together the data analysis and image segmentation components.

III.2.3.2. Concepts and Algorithms Generated

Both the view and controller of the application will be powered by PyQt5, a Python library based on the popular C library Qt. PyQt5 will display any relevant graphics to the user, as well as providing user interaction with the controller. The controller will use PyQt5's event and execution system to keep the application running and responsive to user inputs. It will also handle the flow of data throughout the runtime of the application. This includes importing images, storing data in models for global use, and exporting data and processed images. The view will handle displaying any relevant information to the user. This will include the current state of the program as it processes images and data and visualizations of the results.

III.2.3.3. Interface Description

Services Provided:

Service Name	Service Provided To	Description
Image Importing	Image Segmentation Component	The user will be able to select a folder of images that will be imported into the program. These images will be processed and analyzed by the other components of the application.
Image Exporting	User	Once processed images and data visualizations are available from the data analysis component, the user will have the option to download these files for further use.
Image and Data Visualization	User	Once the processed images and data visualizations are available from the data analysis component, the controller will display these elements along with the rest of the view for quick insight to the user.

Services Required:

Service Name	Service Provided From
Data Visualization	Data Analysis Component

IV. Data Design

Our program utilizes a data structure called a DataFrame from the pandas Python library, which is a 2-dimensional data structure that can act as an array or table like Excel, set with rows and columns. This allows us to store the data measurements obtained from image segmentation (such as size and shape) and tabulate it in a visual and organized way. The filled DataFrame is then used for data manipulation purposes such as creating histograms that illustrate the distributions of each measurement type regarding the pores in each segmented image. However, its further use case can also extend to other features depending on the future needs of the client and our own design choices.

V. User Interface Design

Our current user interface will be a simple design with a focus on ease-of-use and accessibility for the researchers. For now, it will be a simple command line with a menu screen (Image 2) which will list four options: parameters, select images, view results, and close. The parameters button will let the user adjust certain settings and thresholds of the image segmentation process to produce different or more desired results. The select images button will prompt the user to submit a file of images from their local file system to be processed for segmentation. The view results button will display each segmented image and histogram pair on the screen, allowing the user to go through each set individually with buttons. Finally, the close button will shut down the program.

This is more of a placeholder user interface and is not the immediate focus of our team for the program, but instead a task whose attention is relegated more to the second semester of this project. In the future, we plan to implement a more visual and detailed GUI that better resembles a more intuitive application for researchers.

VI. Glossary

Image segmentation - The process of partitioning a digital image into multiple image segments.

VII. References

Software Engineering | Architectural Design, Pranathi. B,
“<https://www.geeksforgeeks.org/software-engineering-architectural-design/>”

VIII. Appendices

Image 1:

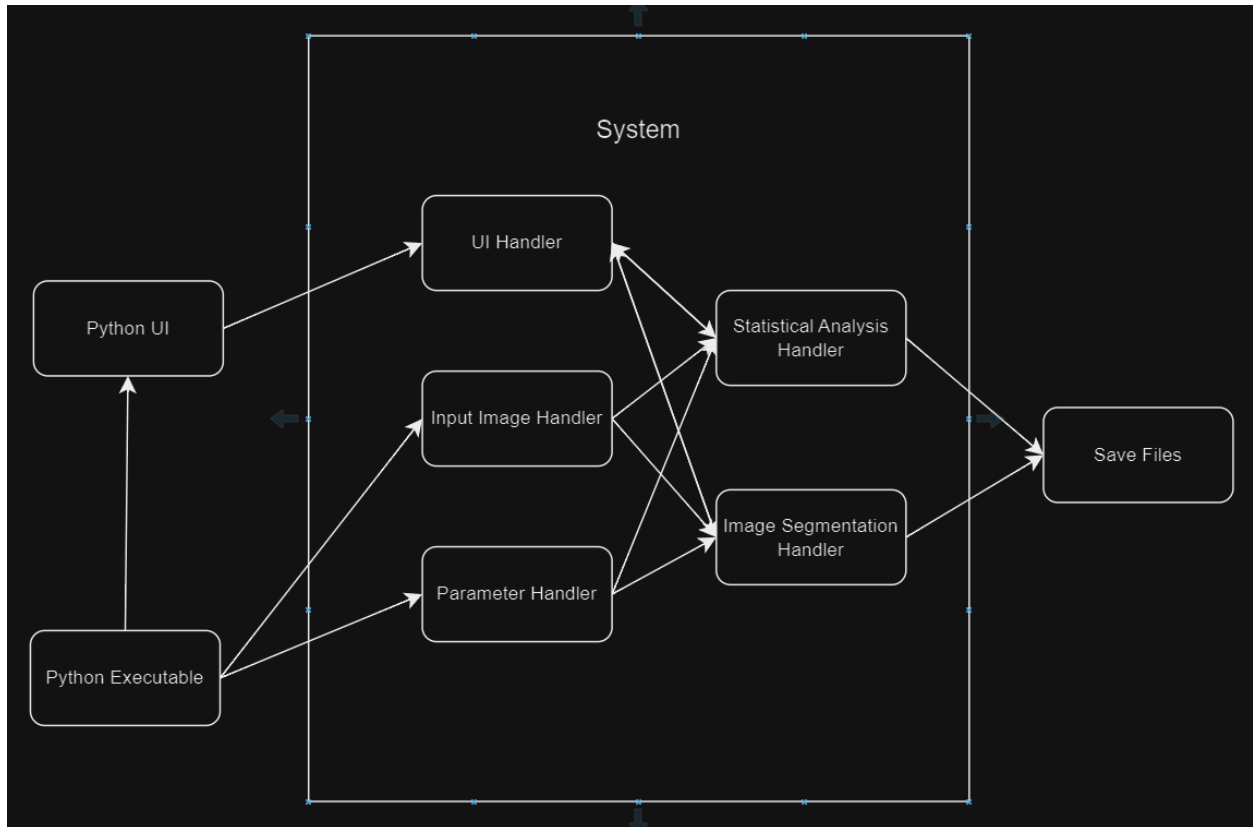


Image 2:

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Parameters

Select Images

View Results

Close